

NIPUN SATHSARA BASNAYAKA

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Associate Software Engineer with hands-on experience developing full-stack applications using React.js, Java (Spring Boot), and Node.js (NestJS). Experienced in building scalable RESTful APIs, developing responsive frontend interfaces, and designing efficient database architectures. Skilled in cloud-based development using AWS, Docker, CI/CD pipelines, and modern backend platforms such as Supabase. Passionate about solving real-world problems through scalable software solutions and collaborating in Agile development teams.

TECHNICAL SKILLS

Programming: TypeScript, JavaScript, Java, SQL

Frontend Development: React.js, Next.js, Angular, Tailwind CSS, Responsive UI Development

Backend Development: Node.js (NestJS, Express.js), Java (Spring Boot), RESTful API Development, Microservices, JWT Authentication, OAuth2

Databases: PostgreSQL, MySQL, MongoDB, Redis

DevOps & Cloud: AWS (EC2, S3), Docker, GitHub Actions, CI/CD Pipelines

Architecture & Design: Microservices Architecture, Event-Driven Systems (Kafka), Clean Architecture, Layered Architecture

Development Practices: Agile/Scrum, Test-Driven Development (TDD), API Documentation (Swagger/OpenAPI), Database Optimization

Third-Party Integrations: Supabase, Google Maps API, Google Sheets API, Firebase

EDUCATION

- **Bachelor of Information Technology**
University of Colombo School of Computing
Aug 2022 – July 2026 (Expected)
- **Diploma in Software Engineering**
Institute of Computer Engineering Technology (iCET)
Aug 2024 – Apr 2025
- **Diploma in Cybersecurity and Ethical Hacking**
SITC Campus
Aug 2022 – Feb 2023

EXPERIENCE

Associate Software Engineer (Full Stack)

Residue Solutions (Pvt) Ltd – Colombo, Sri Lanka

Jun 2025 – Present

- Designed and documented RESTful APIs using Swagger/OpenAPI to support scalable frontend and mobile integrations
- Developed responsive frontend features using React.js and Next.js with efficient API integration and Tailwind CSS
- Built modular backend services using NestJS, Express.js, and TypeScript, ensuring maintainable and scalable architectures
- Optimized PostgreSQL and MongoDB databases using indexing and aggregation queries to improve performance and scalability
- Implemented secure authentication systems using JWT and OAuth2 with Spring Security and NestJS middleware
- Integrated third-party services (Google Maps API, Google Sheets API) to enable location-based features and workflow automation
- Containerized applications using Docker and implemented CI/CD pipelines with GitHub Actions for streamlined deployments
- Diagnosed and resolved production issues related to APIs, integrations, and database performance, improving system reliability

Software Engineering Intern (Full Stack)

Residue Solutions (Pvt) Ltd – Colombo, Sri Lanka

Apr 2025 – Jun 2025

- Built backend APIs using NestJS and Spring Boot for production-ready applications
- Implemented secure authentication systems including JWT-based access control and RBAC mechanisms
- Integrated frontend applications with backend APIs and Firebase services for real-time updates
- Improved application performance through SQL query optimization and structured backend testing using Jest and JUnit
- Collaborated with development teams through Git workflows, pull requests, and code reviews

Software Engineering Trainee

Institute of Computer Engineering Technology (iCET)

Jul 2024 – Apr 2025

- Developed full-stack applications using Java, Spring Boot, Angular, and REST APIs
- Implemented authentication systems including JWT tokens and role-based access control (RBAC)
- Worked within Agile teams using Git version control and collaborative development workflows

PROJECTS

WorkNest – Multi-Tenant Workforce Platform – React.js | Spring Boot | MySQL [GitHub]

- Designed and developed a multi-tenant SaaS platform supporting employee, team, project, task, and attendance management across multiple organizations
- Implemented secure authentication and authorization using Spring Security with JWT and OAuth2, including tenant-based data isolation and role-based access control (RBAC)
- Built scalable backend services and optimized PostgreSQL database schemas, improving system performance, data consistency, and multi-tenant scalability
- Engineered a real-time chat system and task workflows, resolving API performance issues and ensuring reliable synchronization between frontend and backend

Residue ERP System – React.js | Spring Boot | PostgreSQL | Docker

- Developed a full-stack ERP system to manage invoicing, inventory, customers, suppliers, and financial operations with modular, scalable architecture
- Built secure backend services using Spring Boot, JWT authentication, and Spring Security, including role-based access control and API documentation with Swagger
- Designed a feature-driven React frontend using React Query, React Hook Form, and Material UI, enabling efficient state management and responsive user experience
- Containerized and deployed the system using Docker, implementing robust API communication, validation, and error handling for production-ready performance

Task Management System – Next.js + NestJS + MongoDB [Live]

- Built a secure full-stack task management system using Next.js 14 and NestJS
- Implemented JWT authentication with access & refresh tokens via HttpOnly cookies
- Enforced task ownership using JWT guards and protected routes with Next.js middleware
- Applied DTO validation, rate limiting, secure CORS configuration, and environment-based deployment
- Deployed frontend on Vercel and backend services with environment-based configuration for production deployment

Automotive ERP System – NestJS + React.js + MongoDB + AWS [GitHub]

- Developed full-stack features to manage automotive services and internal business operations
- Built secure RESTful APIs for project, task, and user management
- Integrated Firebase for real-time updates and deployed backend services on AWS for scalable cloud-based deployment

EvoPlan Enterprise Web Application – Angular + Spring Boot + OAuth2

- Led backend development and authentication implementation for an enterprise planning system
- Implemented OAuth2-based authentication and Supabase storage integration for file and data management
- Collaborated with 45+ developers using Agile methodology and Git version control

HR Management System – Angular + Spring Boot + MySQL [GitHub]

- Developed modular APIs for employee, payroll, and leave management
- Implemented JWT-based authentication with role-based access control
- Designed a maintainable, scalable architecture with optimized performance

Event-Driven Microservices System – Spring Boot + Kafka + Docker [GitHub]

- Engineered event-driven microservices using Kafka for asynchronous communication
- Designed distributed services for scalable data processing
- Containerized services with Docker to simplify deployment

CERTIFICATIONS

- Docker Foundations Professional Certificate – Docker, Inc. (2025)
- Java Foundations Professional Certificate – JetBrains (2025)
- Postman API Fundamentals Student Expert – Postman (2025)
- Diploma in Cybersecurity and Ethical Hacking – SITC Campus (Top 5, A+)
- AI/ML Engineer – Stage 1 – SLIIT (2025)

ADDITIONAL

Currently preparing for the AWS Certified Solutions Architect – Associate certification.